Karary University

College of Computer science and Technology

CS , Network , IT and SW Engineering

Third Level

INTERNET APPLICATIONS

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المفردات

- Intro to Networking
- Intro to Internet
- E-Mail services
- Browser Server (FTP and HTTP)
- Usenet and Protocol
- Firewall Concepts
- Conference Concepts

GRADE

- **ASSIGNMENTS 20 %**
- **FINAL EXAM 80 %**

Lecture 1

Overview

- Introduction to networks.
- Need for networks.
- Classification of networks.

Introduction to Networks

- A network consists of two or more entities or objects sharing resources and information.
- A computer network consists of two or more computing devices connected to each other to share resources and information.
- The network becomes a powerful tool when computers communicate and share resources with other computers on the same network or entirely distinct networks.

Introduction to Networks

- Computers on a network can act as a client or a server.
- A client is a computer that requests for resources.
- A server is a computer that controls and provides access to resources.

Introduction to Networks

- Data is a piece of information.
- The computing concept ‘hierarchy of data’ is used when planning a network.
- It is essential to maintain a hierarchy of data to manage and control resources among computers.
- Network access to data must be evaluated carefully to avoid security issues.

Need for Networks

- A computer that operates independently from other computers is called a stand-alone computer.
- The process of printing or transferring data from one system to another using various storage devices is called sneakernet.

Need for Networks

- Enhance communication.
- Share resources.
- Facilitate centralized management.

Enhance Communication

- Computer networks use electronic mail (e-mail) as the choice for most of the communication.
- By using networks, information can be sent to a larger audience in an extremely fast and efficient manner.

Share Resources

- A copy of data or application stored at a single central location is shared over a network.
- Computer peripheral devices, referred to as additional components, can be attached to a computer and be shared in a network.

Share Resources

- Peripheral devices include faxes, modems, scanners, plotters, and any other device that connects to the computers.
- Equipment's having common requirements can be shared in order to reduce maintenance cost.

Share Resources

- Important data can also be stored centrally to make it accessible to users, thereby saving storage space on individual computers.
- Computer applications, which take up a considerable amount of storage space, can be installed centrally on the network, saving storage space.

Facilitate Centralized Management

- Networks are used to assist in management tasks associated with their own operation and maintenance.
- Using networks results in increased efficiency and a resultant reduction in maintenance costs.

Facilitate Centralized Management

Software:

- Software is a set of instructions or programs that control the operation of a computer.
- Software can be installed at a central location using servers, where the installation files are made accessible over the network.

Classification of Networks

- Classification by network geography.
- Classification by component roles.

Classification by Network Geography

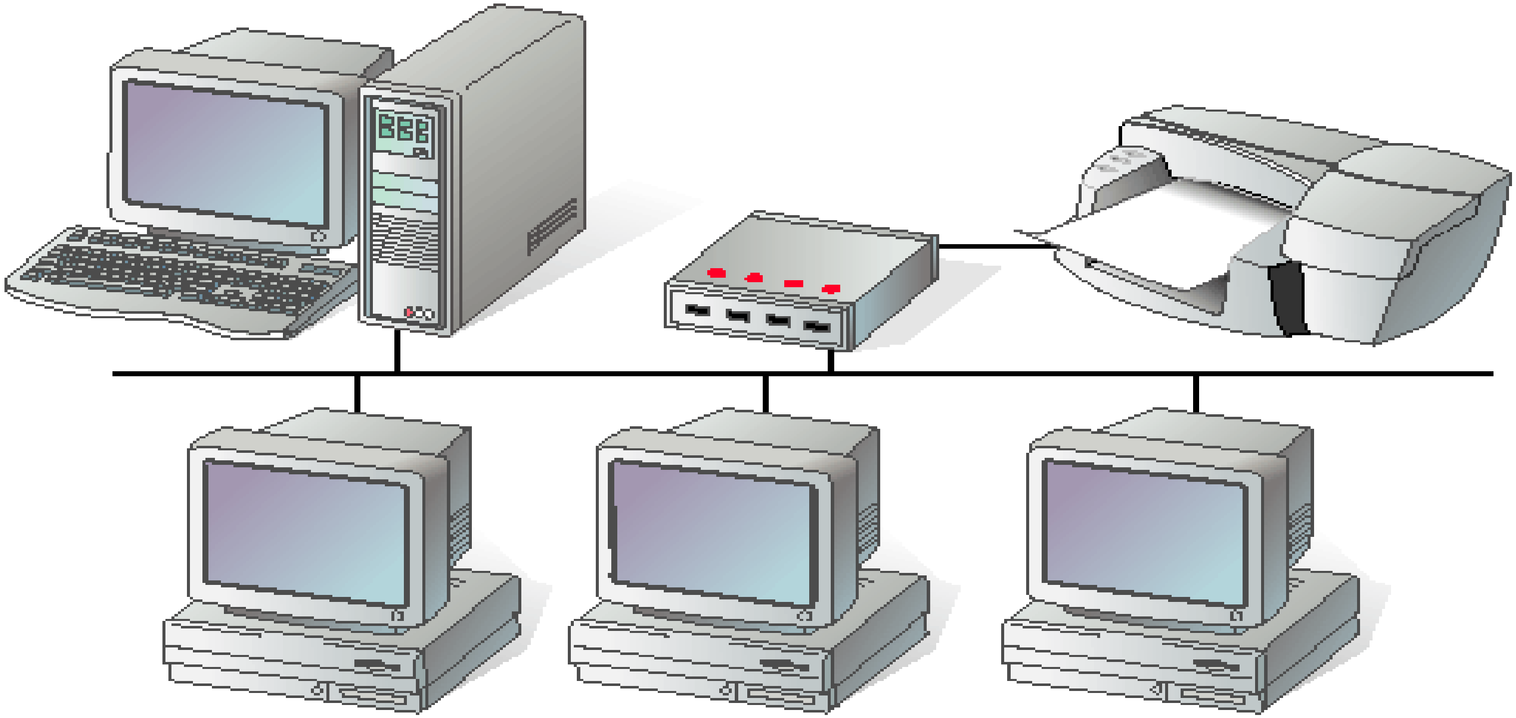
- Networks are frequently classified according to the geographical boundaries spanned by the network itself.
- LAN, WAN, and MAN are the basic types of classification, of which LAN and WAN are frequently used.

Classification by Network Geography

Local area network (LAN):

- A LAN covers a relatively small area such as a classroom, school, or a single building.
- LANs are inexpensive to install and also provide higher speeds.

Classification by Network Geography



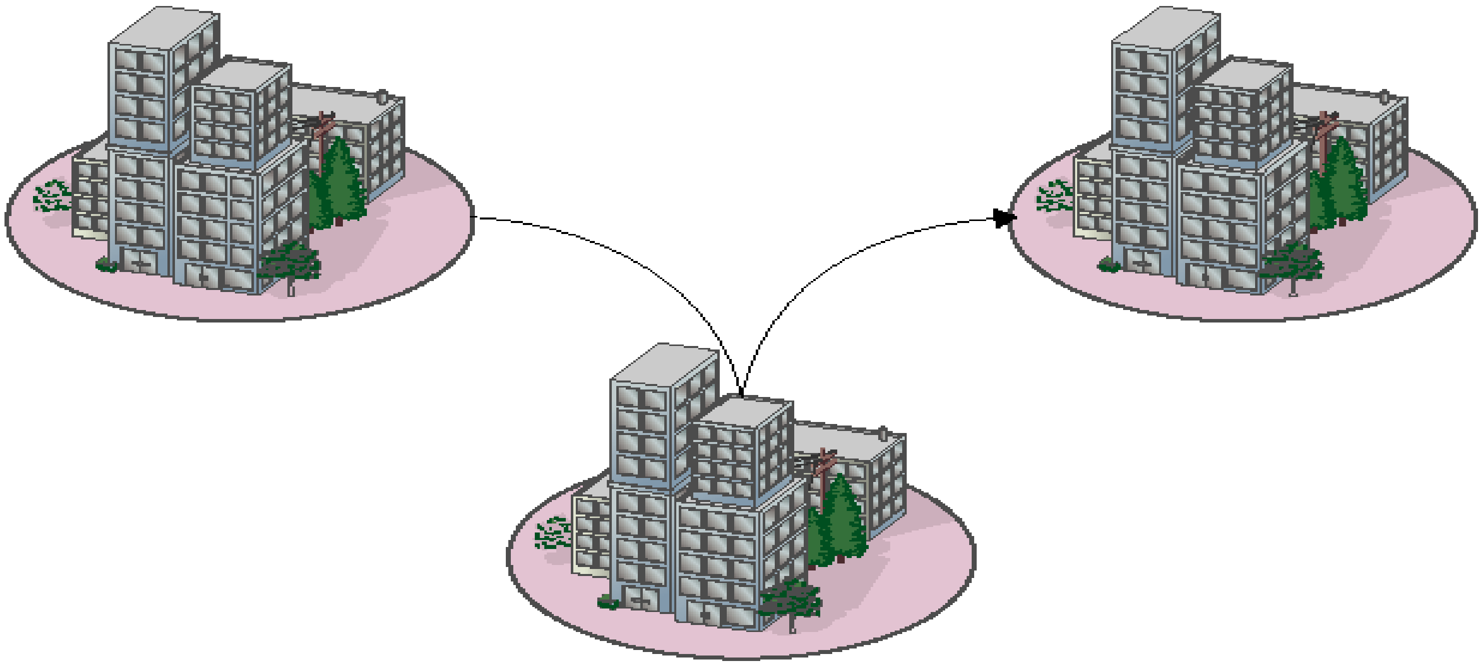
Local area network

Classification by Network Geography

Metropolitan area network (MAN):

- A MAN spans the distance of a typical metropolitan city.
- The cost of installation and operation is higher.
- MANs use high-speed connections such as fiber optics to achieve higher speeds.

Classification by Network Geography



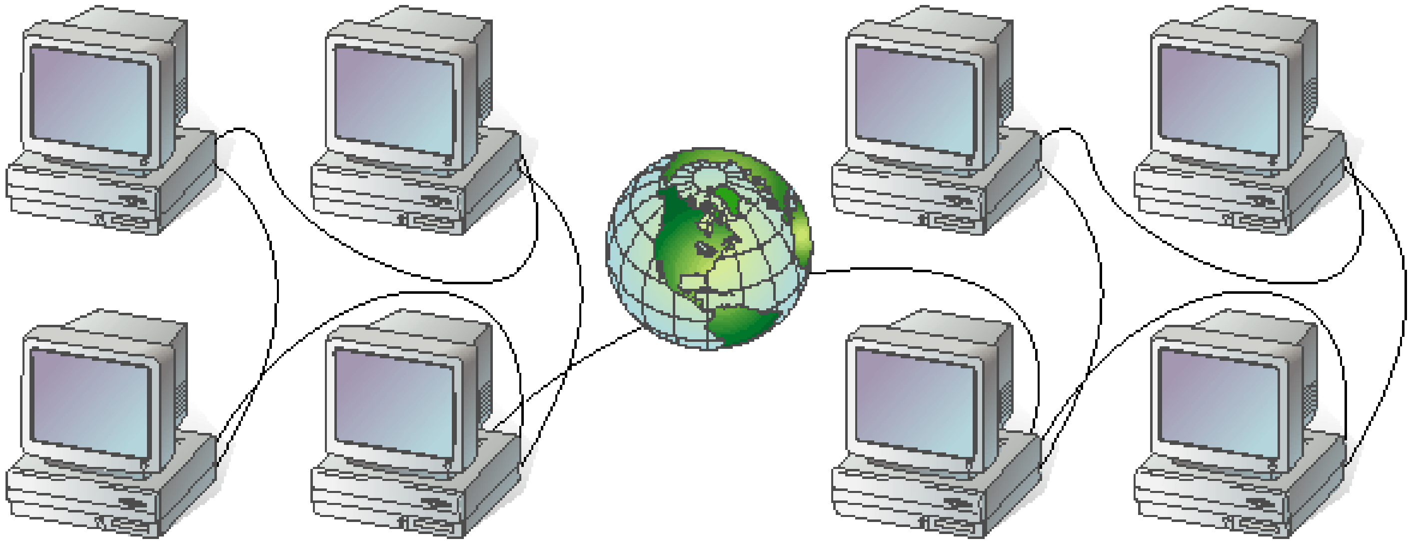
Metropolitan area network

Classification by Network Geography

Wide area network (WAN):

- WANs span a larger area than a single city.
- These use long distance telecommunication networks for connection, thereby increasing the cost.
- The Internet is a good example of a WAN.

Classification by Network Geography



Wide area network

Classification by Component Roles

- Networks can also be classified according to the roles that the networked computers play in the network's operation.
- Peer-to-peer, server-based, and client-based are the types of roles into which networks are classified.

Classification by Component Roles

Peer-to-peer:

- In a peer-to-peer network, all computers are considered equal.
- Each computer controls its own information and is capable of functioning as either a client or a server depending upon the requirement.
- Peer-to-peer networks are inexpensive and easy to install.
- They are popular as home networks and for use in small companies.

Classification by Component Roles

Peer-to-peer (continued):

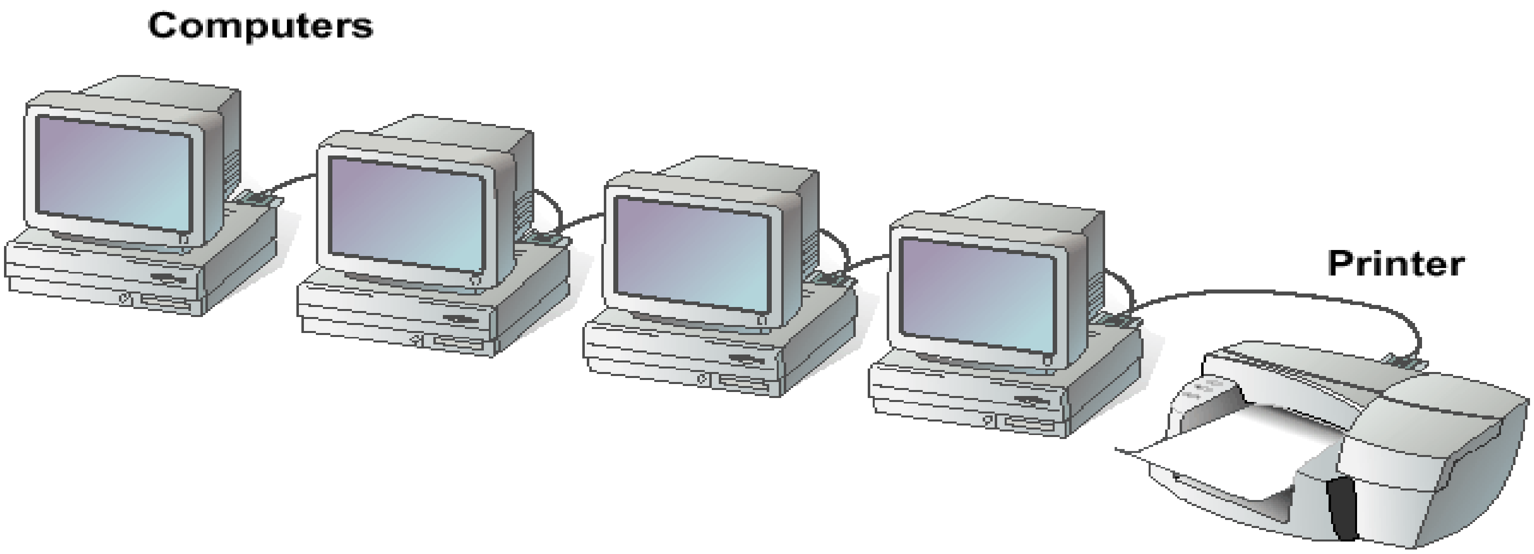
- Most operating systems come with built-in peer-to-peer networking capability.
- The maximum number of peers that can operate on a peer-to-peer network is ten.
- Each peer shares resources and allows others open access to them.

Classification by Component Roles

Peer-to-peer (continued):

- Peer-to-peer networks become difficult to manage when more security is added to resources, since the users control their security by password-protecting shares.
- Shares can be document folders, printers, peripherals, and any other resource that they control on their computers.

Classification by Component Roles



Peer-to-peer network

Classification by Component Roles

Server-based:

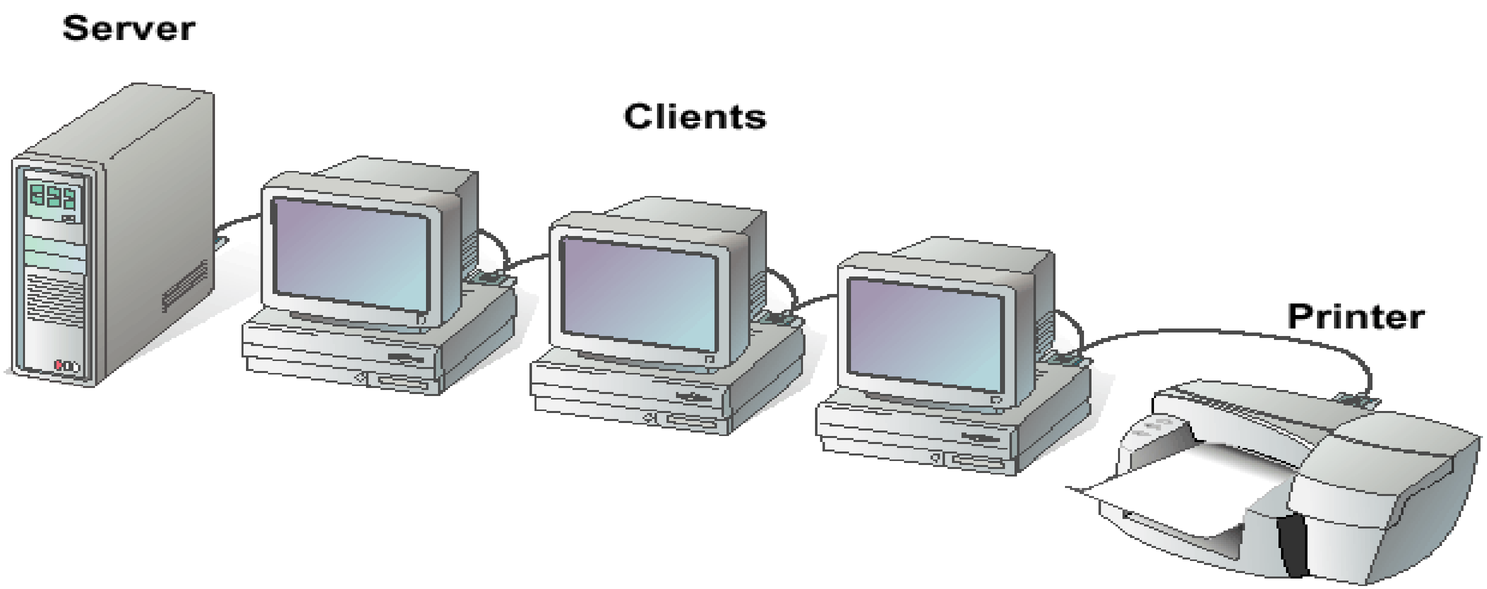
- A server-based network offers centralized control and is designed for secure operations.
- In a server-based network, a dedicated server controls the network.

Classification by Component Roles

Server-based (continued):

- A dedicated server is one that services the network by storing data, applications, resources, and also provides access to resources required by the client.
- These servers can also control the network's security from one centralized location or share it with other specially configured servers.

Classification by Component Roles



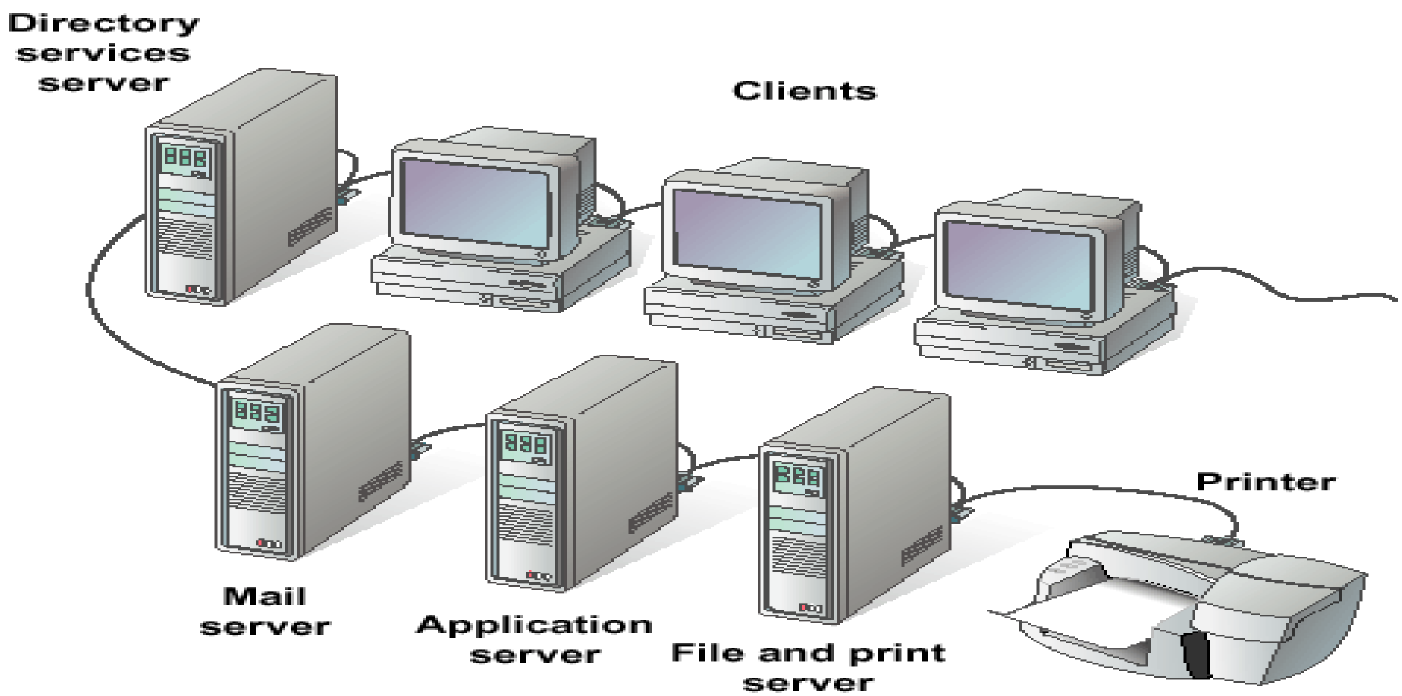
Server-based network

Classification by Component Roles

Client-based:

- Client-based network servers process requests from clients and return just the results.
- These networks take advantage of the powerful processing capabilities of both the client and the server.
- Application servers and communications servers are examples of client-based networks.

Classification by Component Roles



Client-based network

Summary

- A network consists of two or more entities sharing resources and information.
- A computer network consists of two or more computers that are connected and are able to communicate.

Summary

- The basic purpose of networks is to enable effective communication, share resources, and facilitate centralized management of data.
- Networks can be classified according to their geographical boundaries or their component roles.

Questions